UNIBRANCH POINTS OF DECENT SPACES: FOUR EQUIVALENT FORMULATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5690-5700, immediately following the statement at 5674-5688.

Editorial note. Two cross-chapter pointers (More on Algebra, More on Morphisms) and one same-chapter definitional discussion are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Minimal primes versus irreducible components through a point. standard commutative-algebra correspondence invoked at proof lines 5690-5692 (irreducible components of U through u correspond 1-1 to minimal primes of O_{U,u}). Carries the equivalence (1) <=> (2).
- The henselian local ring is the henselization of the chart. discussion following Definition definition-henselian-local-ring of this chapter, cited at proof lines 5693-5695. Connects O^h_{X,x} with the local rings of the elementary etale charts, letting the scheme-side unibranch criterion speak about X.
- Unibranch characterization via henselization. More on Algebra, Lemma more-algebra-lemma-unibranch, cited at proof lines 5696-5697. Carries the equivalence (3) <=> (4).
- Branches count via henselization. More on Morphisms, Lemma more-morphisms-lemma-nr-branches, cited at proof lines 5698-5700. Carries the equivalence (2) <=> (3).

Source reference. Original definition 0BGW: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2716–2716. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BQ0: focused-seeds/geometry/S9-decent-irreducible-local-ring/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CB4: focused-seeds/geometry/S9-decent-irreducible-local-ring/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Let $x \in |X|$ be a point. The following are equivalent*

1. *for any elementary étale neighbourhood $(U, u) \to (X, x)$ the local ring $\mathcal{O}_{U,u}$ has a unique minimal prime,*
2. *for any elementary étale neighbourhood $(U, u) \to (X, x)$ there is a unique irreducible component of U through u,*
3. *for any elementary étale neighbourhood $(U, u) \to (X, x)$ the local ring $\mathcal{O}_{U,u}$ is unibranch,*
4. *the henselian local ring $\mathcal{O}^h_{X,x}$ has a unique minimal prime.*

Proof. The equivalence of (1) and (2) follows from the fact that irreducible components of U passing through u are in 1-1 correspondence with minimal primes of the local ring of U at u. The ring $\mathcal{O}^h_{X,x}$ is the henselization of $\mathcal{O}_{U,u}$, see discussion following Definition 0BGW. In particular (3) and (4) are equivalent by More on Algebra, Lemma 0BQ0. The equivalence of (2) and (3) follows from More on Morphisms, Lemma 0CB4. □